Introduction to Biosignals

- ECG, EMG, EEG, PPG
- Biomedical Signal Fundamentals

Biosignals – Definition

- • Signals generated by physiological activities
- • Electrical or optical in nature
- **Electrical signals** measure voltage generated by ion movement in tissues.
- **Optical signals** measure changes in light due to blood flow in tissues.

- • Reflect organ and system functions

Why Biosignals Matter

- • Disease diagnosis
- • Continuous monitoring
- • Wearable & remote healthcare
- • Research and AI-based analysis

Classification of Biosignals

- • Electrical: ECG, EMG, EEG
- • Optical: PPG
- • Mechanical
- • Chemical
- • Thermal

ECG – Electrocardiogram

- • Electrical activity of the heart
- • P–QRS–T waveform
- • Amplitude: ~1 mV
- • Frequency: 0.05–150 Hz

ECG – Applications

- • Heart rate & rhythm analysis
- • Arrhythmia detection
- • Cardiac stress testing

EMG – Electromyogram

- • Electrical activity of skeletal muscles
- • Generated during contraction
- • Amplitude: μV–mV
- • Frequency: 20–500 Hz

EMG – Applications

- • Neuromuscular disorder diagnosis
- • Prosthetic & robotic control
- • Sports biomechanics

EEG – Electroencephalogram

- • Electrical activity of the brain
- • Measured from scalp electrodes
- • Amplitude: µV
- • Frequency: 0.5–100 Hz

EEG – Brain Waves

- • Delta (0.5–4 Hz)
- • Theta (4–8 Hz)
- • Alpha (8–13 Hz)
- • Beta (13–30 Hz)
- • Gamma (>30 Hz)

EEG – Applications

- • Epilepsy detection
- • Sleep analysis
- • Brain–computer interface

PPG – Photoplethysmogram

- • Optical measurement of blood volume changes
- • Uses LED & photodetector
- • Non-invasive
- • Used in wearables

PPG – Applications

- • Heart rate monitoring
- • Oxygen saturation (SpO_2)
- • Fitness trackers & smartwatches

Comparison of ECG, EMG, EEG, PPG

- ECG: Heart (Electrical)
- EMG: Muscle (Electrical)
- EEG: Brain (Electrical)
- PPG: Blood flow (Optical)

Signal Challenges

- • Noise & motion artifacts
- • Low amplitude signals
- • Baseline drift
- • Need for filtering

Conclusion

- • Biosignals reveal vital physiological information
- • ECG, EMG, EEG, PPG are core signals
- • Foundation of biomedical instrumentation